

Arsha Niksa

Data Scientist | Applied Mathematics & Physics

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A-Niksa

Professional Experience

Data Scientist — Searchwise

Full-time, On-site

Tehran, Iran

Jul 2026 – Present

- Develop and evaluate data-driven algorithms for language-related systems using performance benchmarks and reproducible experimentation.
- Build data pipelines supporting both research experiments and production-oriented processing.

Selected Research Experience

Symmetry-Invariant Kernels for Molecular Shape Spaces

With Marina Meilă, University of Waterloo and Vector Institute

Remote

Mar 2026 – Present

- Investigate kernels on molecular shape spaces invariant under symmetry-group actions, with applications to molecular-dynamics data.
- Develop representations for comparing molecular configurations while factoring out symmetry-equivalent transformations.

Manifold Cohomology Estimation

With J. Jost and M. Eidi, Max Planck Institute

Remote

Jul 2025 – Feb 2026

- Developed methods to estimate manifold cohomology from sampled point clouds and explored dimension reduction that preserves geometric and topological structure.

Topological Data Analysis

Directed reading under H. Edelsbrunner, ISTA

Remote

Jul 2024 – Mar 2025

- Studied computational topology, persistent homology, persistence modules, and algebraic stability.

Education

Sharif University of Technology

BSc (Double Major), Mathematics and Physics

Tehran, Iran

Oct 2021 – Jul 2026

- GPA: 19.54/20.00; ranked 1st in Mathematics.
- Relevant coursework: Data Structures (20/20), Advanced Programming (20/20), Statistics (20/20), Linear Algebra (20/20), Algebraic Topology (19/20), and Topics in Mathematical Biology (19/20).

Technical Skills

Programming: Java, Python, R, Julia, MATLAB, C++

Methods: Data analysis, machine learning, computational geometry, topological data analysis

Languages: Persian (native), Kurdish (native), English (fluent), German (beginner), French (beginner)

Selected Publications

Persistent Patterns in Eye Movements: A Topological Approach to Emotion Recognition

Jul 2025

Arsha Niksa, Hooman Zare, Ali Shahrabi, Hanieh Hatami, Mohammadreza Razvan — arXiv preprint

Effects of Roughness on Rarefied Gas Flow in a Microfluidic Channel

Sep 2021

Arsha Niksa, Amirreza Soheili, Vahid Bazargan — Proceedings of the 19th Fluid Dynamics Conference, SUT

Selected Honors

Bronze Medal (Rank 3), National University Mathematics Olympiad

2024

Top 0.2%, National University Entrance Exam (STEM Stream)

2021

Gold Medal, Persian Young Physicists' Tournament

2020